Teaching Statement

Qiang Ning

"It takes ten years to grow a tree, but hundreds of years to nurture talents" is an old proverb written by GUAN Zhong, a Prime Minister who helped make his state the wealthiest and strongest state in China around 680 BC. The famous leader in China recognized the importance of strategic vision, sustained efforts and a comprehensive plan in education. Over thousands of years, generations after generations have used this proverb as the golden rule to improve the education system to achieve long-term benefits. Now with machine learning (ML) and artificial intelligence (AI) becoming one of the hottest areas in science and technology, it is a crucial time to spend the greatest effort to build the ML & AI educational system, to nurture talents in the domain and have significant impacts to the entire human knowledge and society. Building such a system can never be a short-term task. My teaching goal is to become the leading person in shaping and perfecting the way we learn, use, and benefit from ML & AI. Now with my six-year experience in teaching and mentoring students and my constant reflection along the way, I am determined and confident about next steps in teaching. In the next, I will briefly summarize my teaching experience, my teaching plan, and my specific methods to achieve my goal.

Teaching experience

I have been very passionate about teaching. I have served as a teaching assistant (TA) for two courses during my graduate study. I was on the "List of Teachers Ranked as Excellent by Their Students" across the university, and also nominated by my students for the "Harold L. Olesen Award for excellence in undergraduate teaching" in the ECE department.

The two courses I have taught are a freshman-level course, *ECE 120: Introduction for Computing*, and a senior-level course, *CS 446: Machine Learning*. Both courses had a large size: ECE 120 had more than 300 students, CS 446 more than 200, and that has given me good experience in managing large courses. For both courses, my responsibilities involved designing homework and exam problems, holding office hours, and holding weekly discussion sessions for 50 students. The weekly discussion sessions, in particular, required me to give lectures to the class, from which I have benefited a lot in forming my own teaching strategy and philosophy. The attendance for those discussion sessions was not mandatory, but my session always had the highest attendance.

I have also had the pleasure of mentoring four undergraduate and five younger graduate students, and all of them have been involved in my publications. It is a great self-satisfaction to see them growing more research capabilities and continue their excellence in top places including UIUC, Carnegie Mellon University, the University of Southern California, the University of Pennsylvania, and the Allen Institute of Artificial Intelligence (AI2).

Teaching plan

I am ready to teach undergraduate- and graduate-level courses on machine learning, artificial intelligence, data science, data structures, programming, algorithms, and bioinformatics. I am also happy to teach any introduction courses on math, computing, and electrical engineering. Based on my expertise, I am planning to teach (or develop a new course for) natural language processing (NLP). This course will cover important NLP techniques, including language models (ranging from classical N-Grams to modern word2vec or BERT embeddings), sequence labeling (ranging from shallow parsing to named entity recognition), syntactic parsing (context-free grammar parsing and dependency parsing), and semantic role labeling. I also plan to discuss popular deep learning techniques for various NLP applications such as event understanding, machine translation, question answering, and dialog generation. The course will require a basic understanding of machine learning, linear algebra, and probability theory, and will finally teach students via hand-written assignments, machine problems, and course projects. My goal is to give students an understanding of the difficulty and development of NLP, let them appreciate why these NLP techniques are important in various applications, and finally, cultivate students' interest in pursuing a career in AI.

Teaching methods

Over the past forty years, ML & AI has experienced ups and downs. One of the key issues is the lack of a well-developed education system for ML & AI. As a result, in its good time, lots of practitioners cannot even find a systematic training program, while in its bad time, students lose interest even though it is important. Therefore, it is crucial to develop a good curriculum and teaching strategy to help ML & AI skip the hype cycles but become a powerful tool in tackling a great variety of challenging problems.

1. **Curriculum for ML & AI.** ML & AI are becoming ubiquitous like "programming" in applications ranging from smart homes to autonomous systems, and to general data analysis. As for "programming", over the past decades, we have developed a great curriculum to cover courses from training practitioners to write programs (C programming, data structure, etc.) to training experts to analyze programs (compilers, program verification and synthesis, etc.,). However, we are not doing a perfect job for ML & AI at present: many students learned ML only to find out that all they were able to do was to play with the parameters or structures of neural nets, and some basic concepts like Bayes and data processing were covered repeatedly in different versions of ML classes. I plan to help develop a systematic and comprehensive curriculum for ML & AI: We need to design ML oriented math classes to cover necessary knowledge on linear algebra and probability theory, as well as advanced topics on how to design and investigate ML & AI systems; we need hackathon type of classes to train practitioners to solve real-world problems, and also deep theoretical classes to train experts to make fundamental contributions. Each class should set the right expectations and let students make the most of it.

2. **Be aware of ethics in AI.** People have got excited mostly about the rapid growth and potential impact of AI nowadays, but one less known fact is that the wide usage of ML & AI systems may also cause ethics issues. For example, are the news ranking algorithms creating filter bubbles? How does a car distribute the risk between passengers and pedestrians? Are trading algorithms manipulating the markets? Many police stations are using AI to decide which areas to parole. Will that discriminate against certain racial groups? How to solve such problems may or may not be a computer science issue, but before AI is pervasive in our life, we need to first educate people, especially our students, the potential existence of such issues. This is undoubtedly an important topic to add to our curriculum on ML & AI.

3. **Reinforce the thinking process.** Scientific discoveries or advancements are not always accidental; instead, many of them often come naturally when people ask the right questions. In my lectures, I have always focused on the rationale behind those algorithms and theories. For example, when we were learning linear classifiers, I tried to motivate my students to think more about why we need them, when they are optimal, and how to define "optimality"; when we learned naive Bayes, I tried to motivate them to think about the connection between machine learning and estimation theory, why the generative model is seemingly good, and what other generative models may look like. In my discussion sections, I spent a large amount of time reordering the questions and preparing handouts beforehand with step by step hints as a reflection of the thinking process, so we can always efficiently cover much more contents. I always adopted new teaching techniques such as Jupyter notebook as an interactive tool that could easily switch between theory and programs to reinforce students' computational thinking. Those were the secrets that made my sessions the most popular one among all TAs. This was also why I was nominated for teaching awards multiple times. Constantly improving my teaching skill is and will be an important part of my professional life.

As a researcher in machine learning and artificial intelligence, I feel a strong responsibility to educate the next generation of machine learning scientists, which requires a strategic plan and a strong passion for teaching. My past experience of teaching and mentoring has helped me form my unique teaching strategy and philosophy, and I feel very encouraged by the fact that my students love going to my classes and collaborating with me. I wholeheartedly look forward to continuing my journey and success as a faculty member in your department.